

Intersections of AI, Bots, and Reddit

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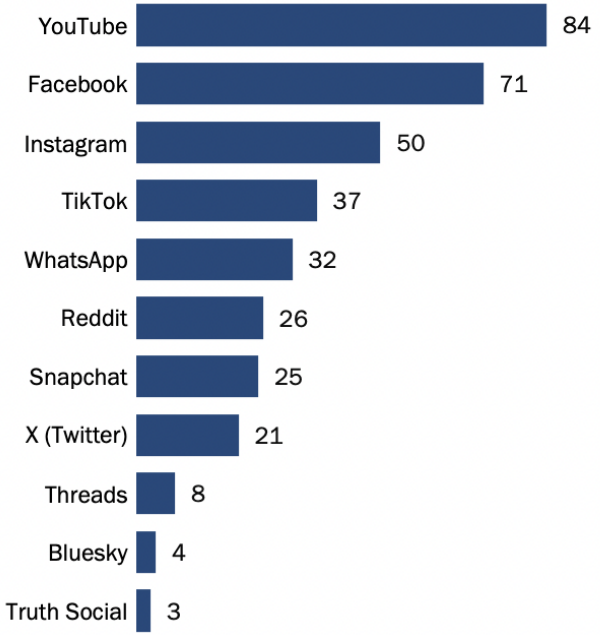
CSE 5825 - BML

Motivation/Background

Social Media/Internet Prevalence

Most U.S. adults use YouTube, Facebook; half report using Instagram

*% of U.S. adults who say they **ever** use ...*



Note: Those who did not give an answer are not shown. Refer to the questionnaire for full question wording.
Source: Survey of U.S. adults conducted Feb. 5-June 18, 2025.
"Americans' Social Media Use 2025"

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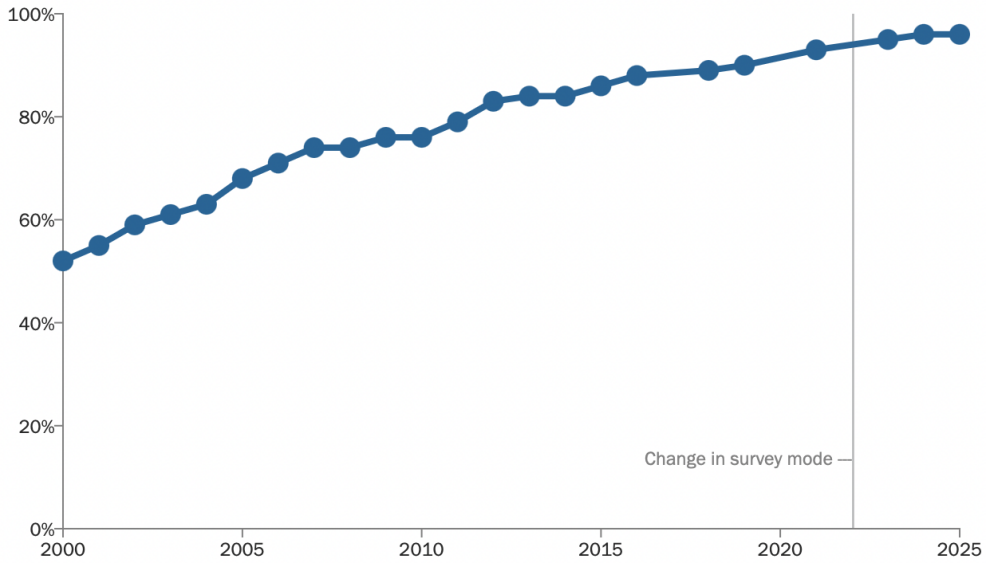
Half of adults say they use Instagram, making it the only other platform in our survey used by at least 50% of Americans.

Smaller shares use the other sites and apps we asked about, such as **TikTok** (37%) and **WhatsApp** (32%). Somewhat fewer say the same of **Reddit, Snapchat** and **X** (formerly Twitter).

Today, 96% of U.S. adults say they use the internet.

Internet use

% of U.S. adults who say they use the internet



Note: The vertical line indicates a change in mode. Polls from 2000-2021 were conducted via phone. In 2023, the poll was conducted via web and mail. In 2024 and 2025, the poll was conducted via web, mail and phone. For more on the mode shift in 2023, read our Q&A. Refer to the topline for more information on how question wording varied over the years. Respondents who did not give an answer are not shown.
Source: Surveys of U.S. adults conducted 2000-2025. Data for each year is based on a pooled analysis of all surveys conducted during that year.

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How do we know what's real and what's fake?

In the July 2026 snapshot, signs of AI authorship can be found in **over one-third of pages** published after ChatGPT was released. This is in line with other studies that have shown that large shares of recently published pages on the internet were likely written or substantially edited by AI.

Source: https://www.pewresearch.org/data-labs/2026/08/20/how-much-of-the-internet-is-written-with-ai/

AI and bots have officially taken over the internet, report finds

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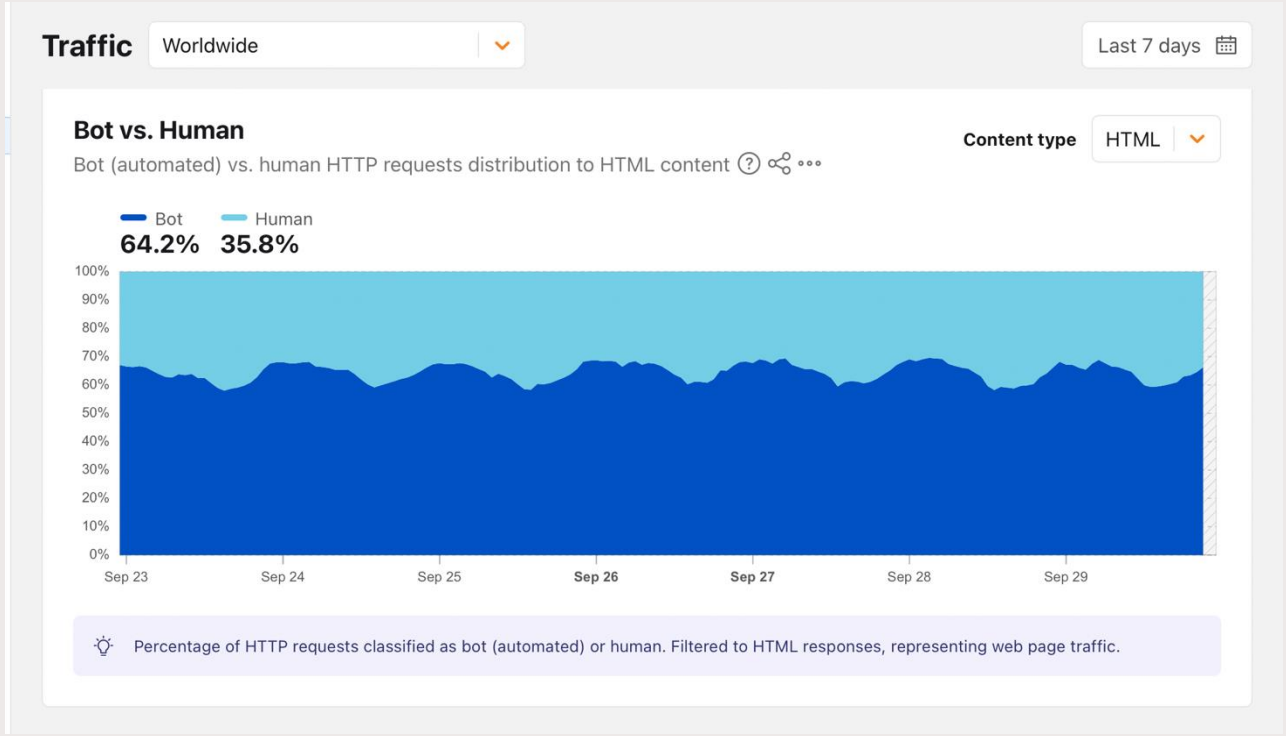
SHARE

"Machine-based traffic is effectively replacing humans as the dominant form of traffic on the other side of the internet,"

Source: https://www.cnbc.com/2026/03/26/ai-bots-humans-internet.html

the internet's history, machines now generate more web traffic than people. Cloudflare's Radar dashboard puts bots at 57.5% of all HTTP requests to HTML content, humans at 42.5%. Prince had predicted the crossover by end of 2027, but we got there eighteen

Source: https://www.forbes.com/sites/josipamajic/2026/06/04/bots-now-outnumber-humans-online-and-the-internet-was-never-built-for-this/



Source: https://radar.cloudflare.com/traffic#bot-vs-human

reddit

Join the most real place on the internet

- Popular, anonymous online forum
- 2025 Pew Research Survey – used by 26% of adults in the US
- Users can:
  - Create posts
  - Write comments
  - Upvote/downvote content
- Subreddit – a subforum on any given topic
  - Governed by its own rules
  - Moderation
  - Community culture

random subreddit 5



r/recruitinghell

Vent your frustrations and laugh (or cry) at absurd job postings.

Weekly visitors

1.4M

View on Reddit

random subreddit 2

r/DIY

Transform your surroundings! Get creative and tackle projects on your own.

Weekly visitors

1.3M

View on Reddit

random subreddit 3



r/TopCharacterTropes

A community to share your favourite character tropes with the community. From characters with similar themes, to weapons with matching vibes.

Weekly visitors

2.7M

View on Reddit

random subreddit 4



r/HistoricalCapsule

Explore fascinating old photos and videos that tell stories from days gone by. Share and discuss about moments that shaped history. Join us in celebrating the beauty of the past, one snapshot at a time.

Weekly visitors

1.8M

View on Reddit

Source: https://www.cnbc.com/2026/03/26/ai-bots-humans-internet.html

Dataset

Dataset

- Huggingface dataset – open-index/arctic
- All Reddit posts and comments since 2005 from every publicly available subreddit
- 1.3 TB, 15.7 billion total entries
  - 12.9 billion comments
  - 2.8 billion posts
- Separate datasets for posts/comments
- Stored as Parquet shards due to size of the dataset

Data Fields

- Reddit-generated ID
- Author
- Subreddit
- Content body
- Title (for posts)
- Timestamp of creation
- Score
  - Number of upvotes/downvotes
  - Can be negative

Comments (`data/comments/YYYY/MM/NNN.parquet`)

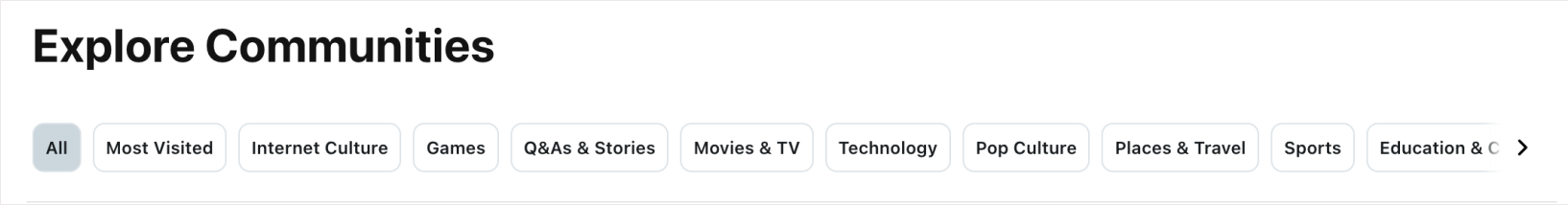
Column	Type	Description
`id`	VARCHAR	Reddit's base-36 comment ID
`author`	VARCHAR	Username. `[deleted]` if account was removed
`subreddit`	VARCHAR	Subreddit name (no `r/` prefix)
`body`	VARCHAR	Comment text in Markdown
`score`	BIGINT	Net upvotes at time of archival
`created_utc`	BIGINT	Unix timestamp
`created_at`	TIMESTAMP	Derived from `created_utc`
`body_length`	BIGINT	Character count of `body`
`link_id`	VARCHAR	Parent submission ID (`t3_...` format)
`parent_id`	VARCHAR	Parent comment or submission ID
`distinguished`	VARCHAR	`moderator`, `admin`, or null
`author_flair_text`	VARCHAR	Author's flair in this subreddit

Submissions (`data/submissions/YYYY/MM/NNN.parquet`)

Column	Type	Description
`id`	VARCHAR	Reddit's base-36 submission ID
`author`	VARCHAR	Username of the poster
`subreddit`	VARCHAR	Subreddit name
`title`	VARCHAR	Post title
`selftext`	VARCHAR	Post body for text posts (empty for link posts)
`score`	BIGINT	Net upvotes at time of archival
`created_utc`	BIGINT	Unix timestamp
`created_at`	TIMESTAMP	Derived from `created_utc`
`title_length`	BIGINT	Character count of `title`
`num_comments`	BIGINT	Comment count on this post
`url`	VARCHAR	External URL for link posts, permalink for text posts
`over_18`	BOOLEAN	NSFW flag
`link_flair_text`	VARCHAR	Post flair text
`author_flair_text`	VARCHAR	Author's flair

Types of Subreddits

- Full list on the Reddit Explore Page



- Can be further narrowed down:

News and Politics

Technology and Science

Business and Finance

Education and Career

Discussion and Support

Media and Entertainment

Lifestyle, Hobbies and Living

Hypotheses

1. Identifying Manufactured Popularity/Astroturfing

Do posts in political, news, technology, and finance subreddits tend to have more abnormal engagement than posts in more casual subreddits such as hobby and general discussion subreddits?

2. Changes in Discourse on AI Over Time

Do technology subreddits display an increase in negative or skeptical language towards AI over time?

3. Detecting AI-Generated Content Across Subreddits

Do certain subreddits contain a higher proportion of posts and comments that appear to be written by AI?

For example, posts with little sentence variety, repetitive phrasing, and overuse of buzzwords.

Thank you!

Questions?